

CSE 403L – Database Management Systems Lab
Department of Computer Systems Engineering
University of Engineering & Technology Peshawar
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Course Description

The objective of this lab course is to cover database modeling, design and implementation and to learn how to create a real-world database application. There are various approaches to designing databases. These approaches are continually evolving. Relational database is emphasized throughout this lab course. Databases are implemented on software packages with an emphasis on SQL.

Week-Wise Course Outline

Week	Contents
Week 1	Introduction to Dynamic Web Content 1. HTML 2. HTTP and HTML 3. The benefits of PHP, MySQL, JavaScript, CSS, and HTML 4. The Apache Web Server 5. About Open Source
Week 2	Static Web Design using HTML 1. Headings 2. Rules 3. Comments 4. Paragraphs 5. Text Formatting 6. HTML Styles 7. Tables 8. Lists 9. Forms DBMS Lab Project: Group Formation 1. Project Group Formation 2. Ideation
Week 3	Improved Web Design using CSS/Bootstrap 1. Font Attributes

	2. Color Attributes 3. Background Attributes 4. Text Attributes 5. Border Attributes 6. Margin Attributes 7. List Attributes 8. User-Defined Selectors (Class & ID) 9. Span and Div. Tags 10. External Style Sheets
Week 4	DBMS Lab Project: Complete Conceptual Schema 1. Entity Description 2. Business Rules 3. Entity Relationship Diagram (ERD) 4. Enhanced Entity Relationship Diagram (EERD – if required)
Week 5	Database Development using MySQL 1. About MySQL 2. Using MySQL Command Line in Windows 3. MySQL Command Categories 4. Use of MySQL Commands
Week 6	Programming in PHP 1. PHP Basics 2. PHP Data Types 3. PHP Expressions 4. PHP Operators 5. PHP Conditionals 6. PHP Loops 7. PHP Functions 8. PHP Advance Features
Week 7	DBMS Lab Project: Normalized Relational Schema & Basic Database 1. Relational Schema 2. 3NF Normalized Relations 3. Tables 4. Metadata 5. Data 6. Constraints
Mid Term Lab Examination	
Week 8	Interfacing PHP with MySQL 1. Type of MySQL Drivers in PHP 2. PHP-MySQL Interfaces 3. Forms representing various entities 4. Forms covering different database operations 5. Login & Signup Information

	6. Session Information 7. Security
Week 9	Session on PHP MVC Framework based Database Application Development
Week 10	Session on Mobile based Database Application Development (Part 1)
Week 11	Session on Mobile based Database Application Development (Part 2)
Week 12	DBMS Lab Project: Complete SQL Database 1. Functions 2. Cursors 3. Stored Procedures 4. Triggers
Week 13	DBMS Lab Project: Database & Front End 1. Forms representing various entities 2. Forms covering different database operations 3. Login & Signup Information 4. Session Information
Week 14	DBMS Lab Project: Complete Project 1. Project Report
Week 15	DBMS Lab Project: Presentations
Final Term Lab Examination	

Course Learning Objectives (CLOs)

CLO #	CLO	Cognitive Domain	PLOs
CLO-1	Use the technology stack needed to develop and deploy database driven application.	C3 (Application)	PLO5 (Modern Tool Usage)
CLO-2	Make user interface using the knowledge of programming languages and software tools to improve user experience.	P4(Mechanism)	PLO5 (Modern Tool Usage)
CLO-3	Apply the knowledge gained from the topics covered in theory and lab to build a complete database driven web and/or mobile application for a real-world problem.	C3 (Application)	PLO3 (Design/ Development of Solutions)

CLOs Assessment Mechanism

Assessment Tools	CLO1	CLO2	CLO3
Lab Reports/Performance	✓	✓	✓
Lab Examination	✓	✓	
Lab Viva			
Semester Project			✓

Grading Criteria

Lab Project & Viva:	50%
Lab Examination:	30%
Hands-On-Experiment/Lab Evaluation/Lab Reports:	20%

Learning Resources for Students

- HTML and CSS Recommended Books:
 1. HTML5 Step by Step, by Faithe Wempen. O'Reilly Media, Inc.
 2. Learn HTML and CSS with W3Schools, by Refsnes & Woodward. Wiley Publishing Inc.
 3. Jump Start CSS, by Louis Lazari. SitePoint Pty. Ltd
 4. Hello! HTML5 and CSS3, by Rob Crowther. Manning Publications
- PHP and MySQL Recommended Books:
 1. Beginning PHP 5.3, by Matt Doyle, Wiley Publishing Inc.
 2. Database Design Manual - Using MySQL with Win, by Matthew Norman, Springer
 3. PHP5 and MySQL Bible, by Converse, Park, & Morgan, Wiley Publishing Inc.
- JavaScript, Ajax, and JQuery Recommended Books:
 1. Jump Start JavaScript, by Pehlivanian & Nguyen, SitePoint Pty. Ltd
 2. Learn JavaScript and Ajax with w3schools, by Refsnes & Henthorne, Wiley Publishing Inc.
 3. JQuery and JavaScript PhraseBook, by Brad Dayley, Addison-Wesley
- SOFTWARE TOOLS:
Sublime Text/Atom/Photoshop/BootStrap/JQuery/WampServer/XAMPP/MySQL

Note: All the lab material will be provided on lab's Google Classroom group.
